Residual Error Estimation and Adaptive FEM

Complete the estimate, then compute and refine

Lecture outline

1. Collect the residual and quasi-interpolation results into the error bound.
2. Compute local indicators and use them in the adaptive loop.
3. Verify the estimator for a localized solution on a square.
4. Additional example: refine near a corner on an L-shaped domain.

Last time we constructed $\mathcal{Q}_h$. Today we use its local approximation estimates to complete the error bound, then use the estimator to guide refinement.

Recall the problem and the error identity

For today's examples, $k = 1$ and all boundary values are zero:

$$-\Delta u = f \text{ in } \Omega, \qquad u = 0 \text{ on } \partial\Omega.$$

Use $\mathcal{V}_0 = H_0^1(\Omega)$ and the conforming P_1 space $\mathcal{V}_{h,0} \subset \mathcal{V}_0$, with

$$a(w, v) = \int_\Omega \nabla w \cdot \nabla v \, \mathrm{d}x, \quad \ell(v) = \int_\Omega fv \, \mathrm{d}x, \quad \|v\|_a = \sqrt{a(v, v)}.$$

Set $e = u - u_h$. Last time, subtracting the FE equation from the continuous equation gave

$$R(u_h; v) = \ell(v) - a(u_h, v) = a(e, v), \qquad R(u_h; v_h) = 0 \quad \forall v_h \in \mathcal{V}_{h,0}.$$

Since $\mathcal{Q}_h e \in \mathcal{V}_{h,0}$, choosing $v = e$ and $v_h = \mathcal{Q}_h e$ gives

$$\boxed{\|e\|_a^2 = R(u_h; e) - R(u_h; \mathcal{Q}_h e) = R(u_h; e - \mathcal{Q}_h e).}$$

Residual representation and local approximation estimates

Write $w = e - \mathcal{Q}_h e$. Elementwise integration by parts gave

$$R(u_h; w) = \sum_{e'} \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x + \sum_{\mathcal{F} \text{ interior}} \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s,$$

where e' indexes elements and $e = u - u_h$ is the error:

$$r_{e'} = f + \Delta u_h, \qquad j_{\mathcal{F}} = -(\nabla u_h^+ \cdot \boldsymbol{n}^+ + \nabla u_h^- \cdot \boldsymbol{n}^-).$$

The normals point outward from the two neighboring elements. To bound w, use the following local approximation estimates for $\mathcal{Q}_h$:

$$\|w\|_{L^2(\mathcal{K}_{e'})} \leq C_Q h_{e'} |e|_{H^1(\omega_{e'})}, \qquad \|w\|_{L^2(\mathcal{F})} \leq C_Q h_{\mathcal{F}}^{1/2} |e|_{H^1(\omega_{\mathcal{F}})}.$$

Here $h_{e'}$ is element diameter, $h_{\mathcal{F}}$ is edge length, and $\omega_{e'}$, $\omega_{\mathcal{F}}$ are their neighboring-element patches. The constant C_Q is independent of mesh size on a shape-regular mesh family.

Mesh-size weights in the residual bound

Start with an element integral in $R(u_h; w)$, with $w = e - \mathcal{Q}_h e$. Cauchy–Schwarz and the preceding local estimate give

$$\left| \int_{\mathcal{K}_{e'}} r_{e'} w \, \mathrm{d}x \right| \le \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \|w\|_{L^2(\mathcal{K}_{e'})} \le C_Q h_{e'} \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} |e|_{H^1(\omega_{e'})}.$$

For an interior edge, the same two steps give

$$\left| \int_{\mathcal{F}} j_{\mathcal{F}} w \, \mathrm{d}s \right| \le \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} \|w\|_{L^2(\mathcal{F})} \le C_Q h_{\mathcal{F}}^{1/2} \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} |e|_{H^1(\omega_{\mathcal{F}})}.$$

These are the origins of the element weight $h_{e'}$ and edge weight $h_{\mathcal{F}}^{1/2}$.

Collect the estimates into the global bound

Add the bounds on the preceding slide, using $\|e\|_a^2 = R(u_h; w)$:

$$\|e\|_a^2 \le C_Q \Bigg[\sum_{e'} h_{e'} \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} |e|_{H^1(\omega_{e'})} + \sum_{\mathcal{F} \text{ interior}} h_{\mathcal{F}}^{1/2} \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})} |e|_{H^1(\omega_{\mathcal{F}})} \Bigg] \le C_Q \eta B.$$

The last step is Cauchy–Schwarz over the combined list of entries, with

$$\eta^2 = \sum_{e'} h_{e'}^2 \|r_{e'}\|_{L^2(\mathcal{K}_{e'})}^2 + \sum_{\mathcal{F} \text{ interior}} h_{\mathcal{F}} \|j_{\mathcal{F}}\|_{L^2(\mathcal{F})}^2,$$

$$B^2 = \sum_{e'} |e|_{H^1(\omega_{e'})}^2 + \sum_{\mathcal{F} \text{ interior}} |e|_{H^1(\omega_{\mathcal{F}})}^2.$$

Complete the reliability bound

Start from $\|e\|_a^2 \le C_Q \eta B$ on the preceding slide. On a shape-regular mesh family, each gradient integral is counted in at most N_{ov} patches. Since $k = 1$,

$$B^2 = \sum_{e'} |e|^2_{H^1(\omega_{e'})} + \sum_{\mathcal{F} \text{ interior}} |e|^2_{H^1(\omega_{\mathcal{F}})} \le N_{\mathrm{ov}} |e|^2_{H^1(\Omega)} = N_{\mathrm{ov}} \|e\|_a^2.$$

Substitute $B \le \sqrt{N_{\mathrm{ov}}} \|e\|_a$:

$$\|e\|_a^2 \le C_Q \sqrt{N_{\mathrm{ov}}} \eta \|e\|_a.$$

Divide by $\|e\|_a$ if it is nonzero:

$$\boxed{\|u - u_h\|_a \le C_{\mathrm{rel}} \eta, \qquad C_{\mathrm{rel}} = C_Q \sqrt{N_{\mathrm{ov}}}.}$$

If the error is zero, the inequality holds without division.

Main result: a computable residual estimator

Assume a bounded Lipschitz domain, $f \in L^2(\Omega)$, conforming shape-regular P_1 meshes, the stated quasi-interpolation estimates, and exact FE assembly and solution. Then

$$\boxed{\|u - u_h\|_a \le C_{\mathrm{rel}}\eta, \qquad \eta^2 = \sum_e \eta_e^2,}$$

where

$$\boxed{\eta_e^2 = h_e^2 \|r_e\|^2_{L^2(\mathcal{K}_e)} + \frac{1}{2} \sum_{\substack{\mathcal{F} \subset \partial \mathcal{K}_e \\ \mathcal{F} \text{ interior}}} h_{\mathcal{F}} \|j_{\mathcal{F}}\|^2_{L^2(\mathcal{F})}.}$$

Each interior edge belongs to two elements, so its total weight is $1/2 + 1/2 = 1$. Summing the local indicators recovers the global formula above.

$$r_e = f + \Delta u_h = f \quad \text{for } P_1 \text{ triangles and } k = 1.$$

For a nonzero source, the volume contribution remains even though $\Delta u_h = 0$ inside an element.

What the estimator tells us

Reliability: $\|u - u_h\|_a \leq C_{\mathrm{rel}}\eta$ bounds the unknown energy error. The constant is independent of mesh size, but need not equal one.

Local indicators: η_e^2 show which elements contribute most to η^2. They are not exact element errors. An efficiency estimate compares them with nearby error and unresolved variation in the data. It requires a separate proof.

Verification with a known solution:

$$E_a = \|u_{\mathrm{ex}} - u_h\|_a, \qquad I_{\mathrm{eff}} = \frac{\eta}{E_a} \quad (E_a > 0).$$

Effectivity is the ratio of the estimator to the actual energy error. Our first example allows this check; the corner example does not.

From the local formula to an assembled vector

Use a discontinuous constant space, called `DG0` in FEniCSx. Its basis function z_e equals 1 on $\mathcal{K}_e$ and 0 elsewhere. The local formula can therefore be written

$$\eta_e^2 = \int_\Omega h^2 r^2 z_e \, \mathrm{d}x + \sum_{\mathcal{F} \text{ interior}} \int_{\mathcal{F}} h_{\mathcal{F}} j_{\mathcal{F}}^2 \frac{z_e^+ + z_e^-}{2} \, \mathrm{d}s.$$

On an edge belonging to $\mathcal{K}_e$, the average of z_e is $1/2$, which assigns the required half contribution to each neighbor.
The notebook assembles volume and facet terms separately and adds them. The resulting vector has one squared indicator per element. No division by element area is needed.

Marking and the adaptive loop

Choose $0 < \theta < 1$. **Dörfler marking** selects a set of elements $\mathcal{M}$ such that

$$\boxed{\sum_{K_e \in \mathcal{M}} \eta_e^2 \geq \theta \sum_{e} \eta_e^2.}$$

Sort squared indicators from largest to smallest and add elements until their sum reaches the target. The notebook uses $\theta = 0.5$.

$$\boxed{\text{solve} \longrightarrow \text{estimate} \longrightarrow \text{mark} \longrightarrow \text{refine}}$$

After refinement, solve the PDE again and recompute all indicators. Neighboring elements may also be refined to maintain a conforming mesh.

An estimator tolerance $\eta \leq \varepsilon_{\text{est}}$ gives $\|u - u_h\|_a \leq C_{\text{rel}} \varepsilon_{\text{est}}$. Today we use fixed cycle counts to keep the demonstration short.

Example 1: a localized solution

Verify the estimator, then compare adaptive and uniform refinement.

A Gaussian pulse on the unit square

Take $\Omega = (0,1)^2$ and prescribe zero values on every boundary edge:

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

Set $\alpha = 400$, $s_i = x_i - \frac{1}{2}$, $G_i = e^{-\alpha s_i^2}$, and $c = e^{-\alpha/4}$. Choose the known solution

$$\boxed{u_{\text{ex}} = (G_1 - c)(G_2 - c)}$$

and define the source by differentiating this expression:

$$f(x_1, x_2) = -\Delta u_{\text{ex}}(x_1, x_2).$$

If $x_i = 0$ or 1, then $G_i = c$, so $u_{\text{ex}} = 0$ on the boundary.
Adapted from Larson–Bengzon, §4.9.3, Eq. (4.130), printed p. 91: normalized and boundary-corrected so the prescribed values hold exactly.

What to examine in the notebook

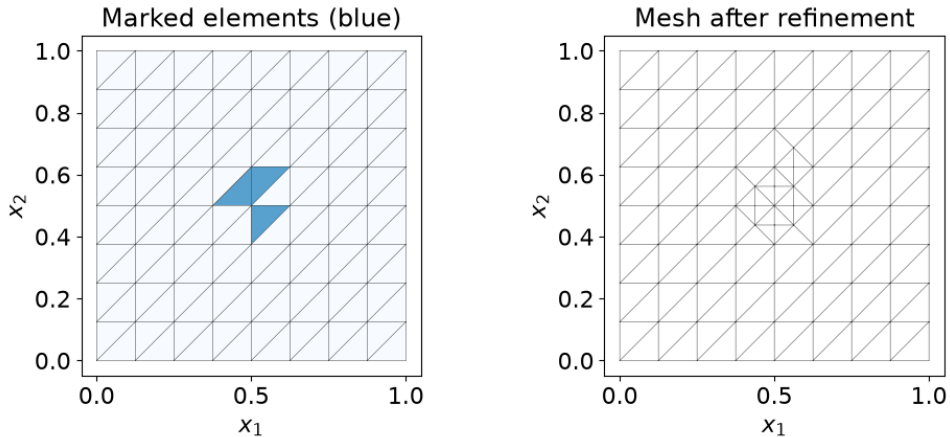
1. Solve on an 8×8 square grid, split into 128 triangles.
2. Plot the FE solution and indicators. Which elements contribute most?
3. Mark and refine once; compute a new FE solution.
4. Repeat and compare with uniform refinement at comparable DOF counts.

Because u_{ex} is known, calculate

$$E_a = \left(\int_\Omega \nabla(u_{\mathrm{ex}} - u_h) \cdot \nabla(u_{\mathrm{ex}} - u_h) \, \mathrm{d}x \right)^{1/2}, \qquad I_{\mathrm{eff}} = \eta / E_a.$$

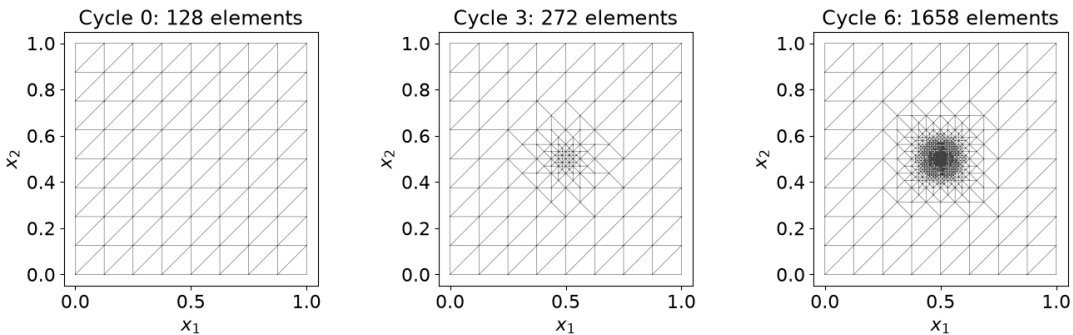
Use the analytic solution, not its nodal interpolant, in this error integral. The Gaussian is not a polynomial: volume quadrature must resolve the pulse.
Open the companion notebook.

One adaptive refinement step



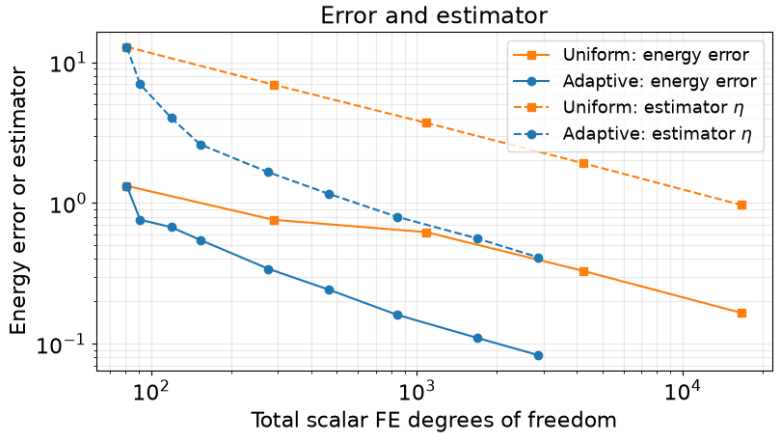
Blue elements are selected from the computed indicators, not from the known solution. The right mesh includes refinement needed for conformity.

Where the mesh becomes finer



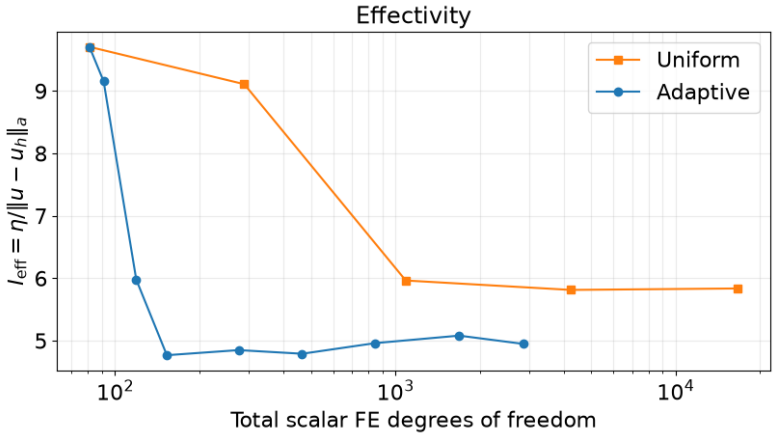
The residual indicators concentrate refinement near the localized pulse. The same solve–estimate–mark–refine operations are repeated at each cycle.

Verify error reduction and estimator behavior



Solid curves: actual energy error. Dashed curves: estimator. Selective refinement gives smaller errors at comparable DOF counts here.

Effectivity: compare the estimator with the actual error

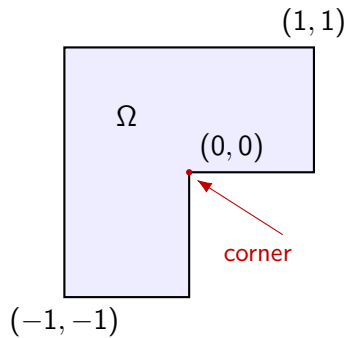


Effectivity is $\eta / \|u_{\text{ex}} - u_h\|_a$. It approaches a factor near five for the adaptive meshes, not one. This ratio is measured here because the exact solution is known.

Additional example: a corner

Use the same estimator when the exact solution is unknown.

Diffusion on the L-shaped domain



Remove the lower-right quadrant:

$$\Omega = \{(x_1, x_2) \in (-1,1)^2 : x_1 < 0 \text{ or } x_2 > 0\}.$$

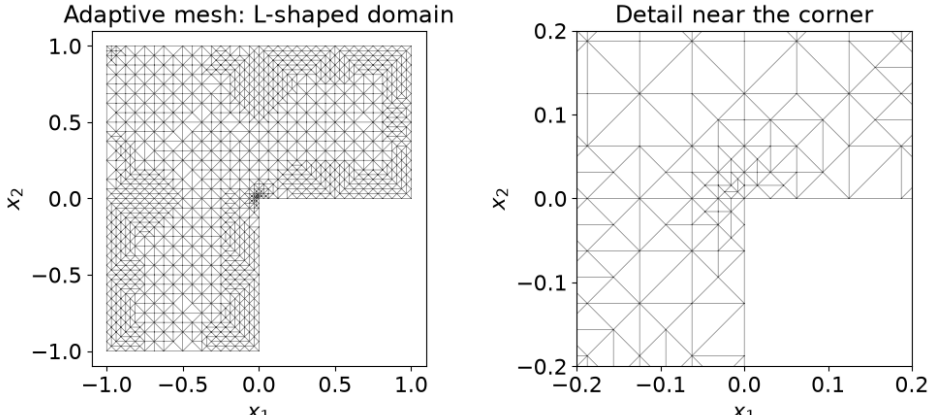
Solve

$$\begin{cases} -\Delta u = 1 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

The same estimator applies, now with $r_e = 1$. The exact solution is unknown, so we cannot compute effectivity here.

Arnold, Chapter 4, §9.5, printed p. 74 (PDF p. 78).

Refinement near the corner



The reentrant corner reduces regularity; the solution is not in $H^2(\Omega)$. Reliability uses the H^1 error, not an H^2 assumption. The indicators produce smaller elements near the corner without a distance-based refinement rule.

Summary

- The Gaussian example compares η with a measured energy error. The actual errors show the benefit of local refinement for this localized solution.
- The corner example shows where the same indicators place resolution when the exact solution is unknown. Its estimator values are not exact errors.
- Both examples retain volume and interior-edge terms and use the same marking and refinement procedure.

$$\|u - u_h\|_a \leq C_{\mathrm{rel}}\eta, \qquad \eta^2 = \sum_e \eta_e^2.$$

References and further details

- Course book, Chapter 8, Sections 8.4–8.5: residual estimator and reliability.
- Larson–Bengzon, 2010 notes, §4.9.1, Theorem 4.7, Eqs. (4.115)–(4.116), printed p. 88; §4.9.3, printed pp. 90–92: adaptive examples, including the Gaussian pulse on p. 91.
- Arnold, *Lecture Notes on Numerical Analysis of Partial Differential Equations*, Chapter 4, §§9.3–9.5, printed pp. 71–76: residual estimates, marking, and the L-shaped example.

The supporting reading decks retain the longer derivations and calculations.